Turning Pre-Trained Vision Transformers into End-to-End Histopathology Whole Slide Image Models for Survival Prediction

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Abstract

Conventional whole slide image (WSI) analysis pipelines follow a two-stage process. First, an image encoder, such as a vision transformer (ViT), is used to perform batched offline feature extraction on a series of tiles cropped from the WSI. Second, a multiple instance learning (MIL) model is trained with slide-level labels to obtain task-specific slide embeddings. However, several limitations exist: strong reliance on pre-trained weights of the tile encoder, the absence of receptive fields from the original image, and a lack of task-independent WSI representations. An ideal improvement would be to develop an end-to-end pre-trained WSI model, but training it from scratch will face challenges such as high training costs and computational complexity. In this work, we deconstruct the key steps of ViT-based pathology image representation and propose a conversion strategy called E2E-ViT, which transforms a vanilla ViT into an end-to-end pre-trained WSI model without introducing additional parameters. E2E-ViT directly inputs the entire tissue region in WSIs to efficiently feed image sequences into the transformer backbone, achieving information interaction from the original receptive fields and generating slide features. Through multiple survival prediction tasks, we demonstrate that transformed pre-trained ViTs outperform two-stage MIL models and slide foundation models (SFM). Our work presents a new end-to-end learning paradigm that provides a promising direction for the next generation of computational pathology models.

1. Introduction

By comprehensively digitizing histopathological slides at gigapixel resolution, the WSI has become an indispensable component of modern clinical workflows, opening up unprecedented opportunities for computational analysis [1, 29, 43]. However, the huge size, high heterogeneity, and lack of pixel-level fine annotation in WSIs pose significant challenges to traditional supervised learning [16, 40]. Therefore, current research has shifted toward pre-training paradigms, such as models with self-supervised learning (SSL) [6, 20, 44, 48], to build transferable representations that can generalize across diseases, organs, and tasks in computational pathology [3, 30, 40].

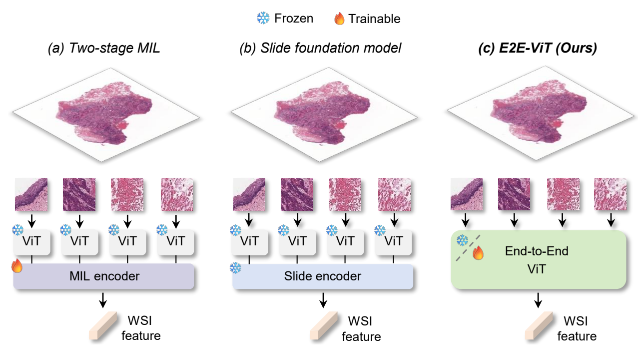

Figure 1. Illustration of three different computer vision frameworks for WSI analysis. (a) Two-stage MIL framework, which uses a pre-trained ViT to offline encode tile-level features, followed by training a separate MIL encoder. (b) SFM, which uses a pre-trained ViT to offline encode tile-level features, and then employs a pre-trained slide encoder. (c) Our proposed E2E-ViT, which uses a pre-trained ViT directly for WSI offline encoding or further fine-tuning.

Existing pre-trained models primarily focus on feature extraction from tile-level images, and are subsequently used in two-stage WSI analysis pipelines [41, 55]. Typically, a pre-trained tile encoder is used to perform offline encoding of tiles cropped from a WSI. Then, with slide-level labels, a MIL framework is trained to aggregate these tile features into a slide-level prediction. Despite significant progress, this two-stage paradigm has several inherent limitations. First, it heavily relies on pre-trained weights, which are not updated with downstream tasks, thus becoming decoupled

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from the slide-level context. Second, batched offline encoding deprives the ability to directly perceive regional interactions from the information in the original image, resulting in the loss of global receptive fields and spatial continuity, both of which are crucial for modeling histological structure. Third, the generated slide representations are often task-dependent, lacking the generalizability required for downstream adaptation. Although recent SFMs aim to further pre-train a slide encoder on top of pre-trained tile encoders to obtain general slide representations [10, 49, 54], they still discard receptive-field information from the raw image and rely on well-performing tile encoders.

An ideal solution is to develop an end-to-end pre-trained WSI model that directly ingests the entire slide and optimizes global representations within a unified framework. However, training such models from scratch faces significant challenges. First, the computational cost is extremely high. Backpropagating at the native resolution of a single WSI far exceeds the capacity of typical training hardware. Second, the amount of pre-training data at the WSI level remains insufficient. Publicly available WSI datasets typically contain tens of thousands of slides [12, 28, 50]. In contrast, large-scale pre-training in the computer vision community usually relies on millions of images, leaving a substantial gap in data scale. Some previous works have explored end-to-end solutions [33, 35, 42, 53]. However, they either downsample the original images, which compromises the effective receptive field, or adopt a one-task one-weight paradigm, which only produces task-specific slide features.

In fact, a purely from-scratch pipeline is not the only viable path. Pre-trained tile encoders, especially those ViT-based foundation models adapted to pathology images [6, 13, 17, 26, 27, 44], have already learned cross-regional interactions among local patches and exhibit strong extrapolation capabilities at higher resolutions [6]. If such tile-level priors can be smoothly extended to slide-level modeling, theoretically, one can obtain a pre-trained end-to-end WSI model without training from scratch. Inspired by this, in this work, we propose E2E-ViT, a conversion strategy that turns tile-level ViT encoders into WSI models without adding parameters. E2E-ViT makes three key modifications on top of the original architecture. First, for the model input part, we treat the entire WSI as a continuous, long-strip image fed into the ViT, enabling it to be segmented and encoded in non-overlapping convolutional layers, thereby preserving the original spatial organization. Second, for the sequence compression part, since the encoded patch token sequence is extremely long, it significantly increases inference latency. We introduce a patch merger that compresses the sequence while preserving the original receptive fields, enabling efficient feature aggregation. Finally, for the position embedding part, considering that these transformer backbones are usually pre-trained on short sequences, we add relative positional encoding to enhance extrapolation under long-sequence, high-resolution inputs. Figure 1 shows the formal differences between two-stage MIL, SFMs, and proposed E2E-ViT. We perform E2E-ViT conversion and validation on three types of tile encoders: ViT-Small (ImageNet pre-trained) [11], CONCH (vision-language pre-trained) [26], and H0-mini (SSL pre-trained) [13]. Across five survival prediction tasks, E2E-ViT consistently outperforms both MIL-based pipelines and SFMs. We also conduct experiments, such as ablation studies on computational efficiency and adaptability to different-sized models, which demonstrate the feasibility and advantages of the proposed E2E-ViT. In summary, our main contributions are as follows and the code is available at https://github.com/WonderLandxD/E2E-ViT.

1. We analyze representation patterns learned by ViT-based pre-trained encoders on pathology images and reveal that their cross-regional modeling capability can naturally extend to slide-level resolutions.

2. We propose E2E-ViT, a simple yet effective conversion strategy that turns tile encoders to WSI-level modeling without introducing any learnable parameters.

3. We demonstrate that the converted models not only support end-to-end fine-tuning and outperform two-stage MIL methods, but also enable offline WSI feature encoding, achieving superior linear-probe performance over mainstream SFMs.

2. Related works

2.1. Pre-Trained Encoders for Tile-Level Images

Image encoders trained on large-scale tiles can map pathology images into features that are rich in prior information, supporting a variety of downstream tasks. Due to the scarcity of annotations in pathology, early studies predominantly employed ImageNet [9] pre-trained convolutional neural networks or ViTs as backbones for region-of-interest analysis [23, 45, 57] or as feature extractors in two-stage MIL pipelines [25, 39]. With the rise of SSL, modern encoders can be pre-trained on unlabeled pathology images. For example, methods such as SimCLR [7], MoCo [14], and DINO [4] have been used to learn from millions of tiles cropped from hundreds of thousands of WSIs, developing a series of pre-trained models at different parameter scales [20, 22, 48]. Among these, ViT backbones trained under DINOv2 [31] frameworks, such as UNI [6], Prov-GigaPath [54], and Virchow [44], have emerged as widely adopted image encoders in digital pathology. Moreover, several works also explore image–text contrastive ViT models, such as PLIP [17], CONCH [26], and MUSK [52], which, via frameworks like CLIP [36] and CoCa [58], not only produce strong vision encoders but also enable zero-shot classification, cross-modal retrieval, and caption generation.

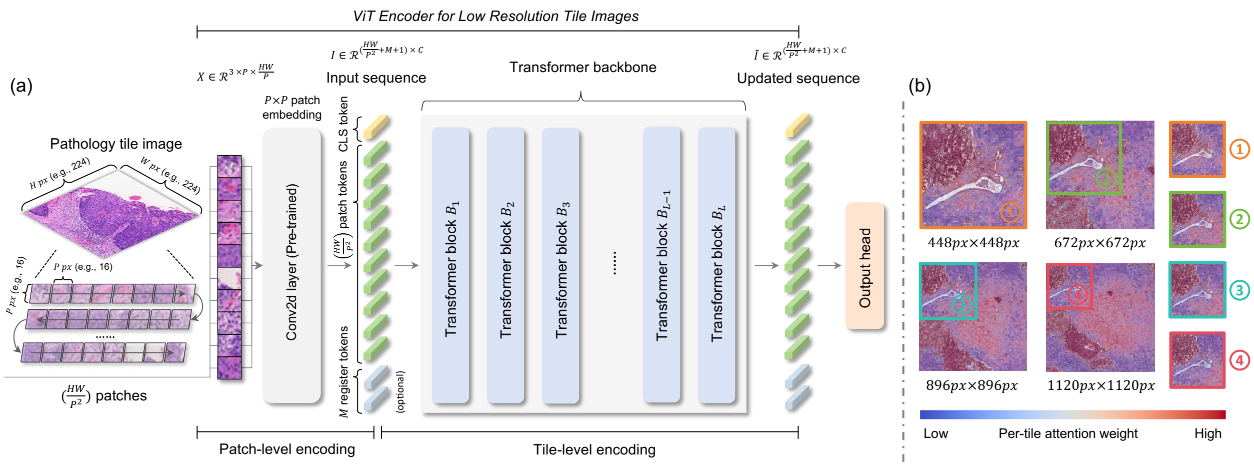

Figure 2. (a) The ViT model framework for low resolution pathology images, which mainly consists of patch-level encoding and a tile-level transformer backbone. (b) Attention heatmaps of a pre-trained ViT in the same field of view at different resolutions in pathology regions.

2.2. Two-Stage MIL in Histopathology WSIs

MIL has long been a core paradigm for WSI analysis. In this setting, features encoded by offline tile encoders are treated as instances, and the entire slide is modeled as a bag, enabling effective downstream prediction [55]. Representative methods such as ABMIL [18] and CLAM [25] learn attention scores over instances and aggregate tile features via attention-weighted pooling to derive slide embeddings. DSMIL [22] further exploits WSI multi-scale characteristics to obtain global embedding. Approaches such as TransMIL [39], WiKG [24], RRTMIL [41], and 2DMamba [60] introduce improvements from the perspectives of transformers, graph modeling, online re-embedding, and state space architecture, respectively. However, these methods require training from scratch for each task, cannot produce task-agnostic slide representations, and remain highly dependent on the capacity of the pre-trained tile encoder.

2.3. WSI Models in Digital Pathology

Most existing WSI models fall under SFMs, which are trained on top of a pre-trained tile encoder. For example, CHIEF [49] uses tile features from CTransPath [48] and combines contrastive learning with anatomical priors to train an attention-based aggregator. MADELEINE [19] performs multi-stain contrastive learning with tile features from CONCH. GigaPath [54] leverages features from Prov-GigaPath to perform the MAE [15] pretext task. FEATHER [38] pre-trains an ABMIL via a 108-class pan-cancer supervised objective. While these models can produce task-agnostic slide embeddings, these models never "see" the raw slide image during training, which discards original receptive-field information and may limit potential gains. Beyond this two-stage pre-training route, earlier studies, such as StreamingCNN [33], DSNet [53], and LongViT [47], explore training directly on resized thumbnails. Recently, approaches such as ABMILX [42] and Pixel-Mamba [35] have been tailored as end-to-end WSI architectures, reporting promising results. However, they lack pathology-domain pre-training weights and thus do not yield transferable slide representations.

3. Methodology

3.1. Revisit ViT in Pathology Images

As shown in Figure 2.(a), ViTs for image representation mainly consist of two modules: a non-overlapping 2D convolutional layer with kernel size P that partitions an image into patch tokens, and a transformer backbone with L layers that performs sequence modeling through self-attention. In most prior studies, images are reshaped to a fixed low resolution (e.g., 224×224) for convenience and efficiency in batching. In fact, the image resolution only needs to satisfy that its height H and width W are multiples of the kernel size P, allowing the image to be tokenized into (HW/P^2) patch tokens and performing forward and backward propagation seamlessly. After tokenization, the ViT adds position embeddings to each patch token and introduces a CLS token, optionally along with register tokens [8]. These tokens are then concatenated into a single sequence and processed by the transformer backbone, where the self-attention transmits information among tokens to produce expressive image representations. In theory, if a pre-trained ViT has effectively learned inter-patch interaction patterns, such priors should remain extrapolatable and continue to function effectively even when applied to longer token sequences derived from higher-resolution inputs.

To evaluate the extrapolation of ViT representations over longer token sequences, we input the four resolution vari-

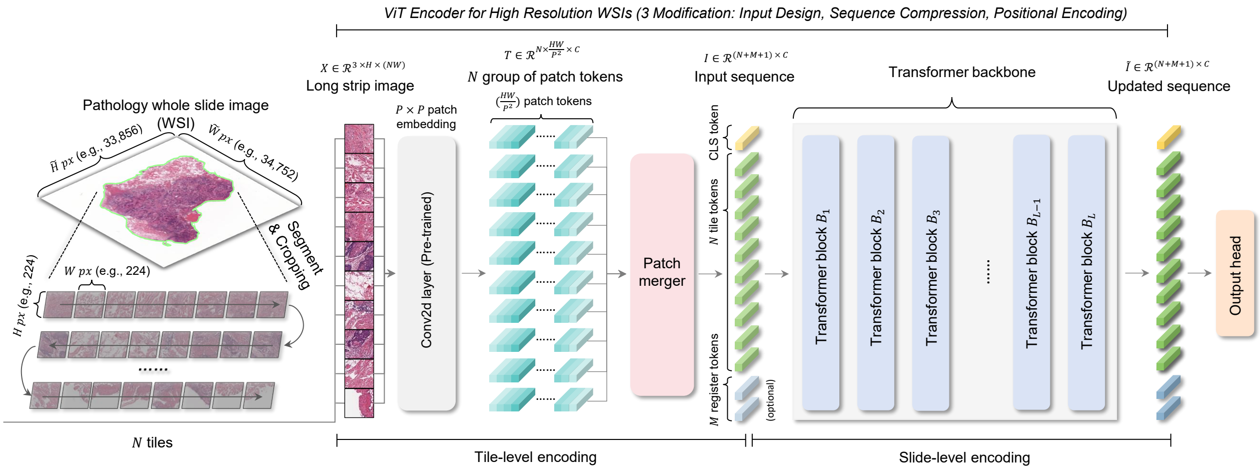

Figure 3. The ViT model framework for high resolution WSIs converted by our proposed E2E-ViT strategies, including tile-level encoding and a slide-level transformer backbone. Three modifications are performed compared to vanilla ViT: input design, sequence compression, and positional encoding.

ants into the same pre-trained ViT model CONCH and visualized the final-layer attention heatmaps of the CLS token. As shown in Figure 2.(b), the attention patterns within the overlapping region remain largely consistent across resolutions, while attention smoothly extends to the newly revealed peripheral areas. This behavior indicates that the pre-trained ViT has learned cross-patch interaction priors that remain effective at higher input resolutions, thus supporting our hypothesis.

3.2. E2E-ViT Conversion Strategy

Building on the above discussion, our proposed E2E-ViT aims to transform a ViT into a model that is operable on higher-resolution WSIs. Specifically, we introduce three modifications to the original ViT: (1) an input design tailored for WSIs, (2) a sequence compression mechanism, and (3) improved positional encoding. Figure 3 illustrates the overall E2E-ViT framework.

3.2.1. Input Design

A straightforward approach is to feed a WSI at a fixed magnification directly into ViT. However, lots of WSIs, such as biopsy slides, contain extensive background that is irrelevant to downstream tasks. We therefore perform preprocessing. Specifically, at a chosen magnification, we obtain a tissue mask using OTSU [32] or GrandQC [51] to remove background, then apply a sliding window to crop non-overlapping tiles of size $Hpx \times Hpx$ (H needs to be chosen as a multiple of the ViT kernel size P). We collect N tiles and concatenate them to form a long strip image of shape $3 \times H \times (HN)$. This input design both excludes task-irrelevant background and exposes all tissue content in the WSI to the model, facilitating subsequent processing.

3.2.2. Sequence Compression

Given that the ViT kernel size is typically small (e.g., 8, 14, and 16) relative to WSI resolution, feeding a long strip image into the patch-embedding layer produces a highly long token sequence of length $N \times (H/P)^2$ with channel dimension C. This makes inference and further fine-tuning computationally impractical. Therefore, for the patch-token set of each tiles $T_i = [t_{i,1}, t_{i,2}, \ldots, t_{i,(H/P)^2}]^\top \in \mathcal{R}^{(H/P)^2 \times C}$, inspired by [46], we introduce a patch-merger module that compresses tokens via mean pooling:

$$I_i = \frac{1}{(H/P)^2} \sum_{k=1}^{(H/P)^2} t_{i,k}. \tag{1}$$

Although simple, this parameter-free merging mechanism preserves the consistency of the tile-token feature space with that of the original patch tokens, allowing for the direct use of pre-trained weights and providing a fine-tuning-friendly interface for downstream tasks.

Notably, instead of merging tokens tile by tile, patch merger can also partition the entire token sequence into G groups $\{\hat{T}_i\}_{i=1}^{G}$ and then perform patch merging within each group:

$$\begin{cases} \{\hat{T}_1, \hat{T}_2, \ldots, \hat{T}_G\} = \text{Split}\big(\{T_1, T_2, \ldots, T_{(H/P)^2}\}\big), \\ \hat{I}_i = \dfrac{1}{|\hat{T}_i|} \displaystyle\sum_{k=1}^{|\hat{T}_i|} \hat{t}_{i,k}, \quad i = 1, 2, \ldots, G. \end{cases} \tag{2}$$

This allows flexible control over the sequence length, enabling adaptation to different downstream task configurations or computational hardware constraints.

Backbone	CCRCC	HNSC	LUAD	PDAC	MBC	Overall
ViT-Small [11] + ABMIL [18]	0.4978 ± 0.0848	0.5659 ± 0.1005	0.5283 ± 0.0309	0.5093 ± 0.1076	0.6774 ± 0.2188	0.5557
ViT-Small [11] + CLAM [25]	0.4830 ± 0.0521	0.5451 ± 0.0762	0.5173 ± 0.0533	0.5150 ± 0.0568	0.4932 ± 0.1775	0.5107
ViT-Small [11] + DSMIL [22]	0.4972 ± 0.0488	0.5857 ± 0.0987	0.5710 ± 0.0535	0.5905 ± 0.0680	0.5129 ± 0.1289	0.5515
ViT-Small [11] + TransMIL [39]	0.5895 ± 0.0278	0.6682 ± 0.0699	0.6372 ± 0.0438	0.6377 ± 0.0852	0.6604 ± 0.1978	0.6386
ViT-Small [11] + WiKG [24]	0.5737 ± 0.0357	0.5959 ± 0.0757	0.5796 ± 0.0428	0.5986 ± 0.0902	0.6834 ± 0.2141	0.6062
ViT-Small [11] + RRTMIL [41]	0.5538 ± 0.0265	0.5818 ± 0.0610	0.6403 ± 0.0692	0.6029 ± 0.1681	0.6835 ± 0.1785	0.6125
ViT-Small [11] + 2DMamba [60]	0.5833 ± 0.0332	0.6677 ± 0.0568	0.6135 ± 0.0487	0.6221 ± 0.1425	0.5198 ± 0.1648	0.6013
E2E ViT-Small (Ours)	**0.5963 ± 0.0801**	**0.7018 ± 0.0499**	**0.6967 ± 0.0601**	**0.6408 ± 0.0671**	**0.6977 ± 0.1631**	**0.6667**
CONCH [26] + ABMIL [18]	0.6109 ± 0.1132	0.6858 ± 0.0327	0.6791 ± 0.0806	0.6249 ± 0.1334	0.5794 ± 0.1650	0.6360
CONCH [26] + CLAM [25]	0.6120 ± 0.1163	0.6921 ± 0.0600	0.6816 ± 0.0652	0.6350 ± 0.1367	0.5849 ± 0.1416	0.6411
CONCH [26] + DSMIL [22]	0.5973 ± 0.1160	0.7299 ± 0.0449	0.7036 ± 0.0621	**0.6535 ± 0.0782**	0.5388 ± 0.1820	0.6447
CONCH [26] + TransMIL [39]	0.6815 ± 0.0366	0.7325 ± 0.0485	0.6865 ± 0.0627	0.6375 ± 0.0646	0.6163 ± 0.0878	0.6709
CONCH [26] + WiKG [24]	0.6762 ± 0.0343	0.7417 ± 0.0579	**0.7116 ± 0.0318**	0.6210 ± 0.0457	0.6971 ± 0.1499	0.6895
CONCH [26] + RRTMIL [41]	0.6747 ± 0.0392	0.7412 ± 0.0406	0.7073 ± 0.0323	0.6403 ± 0.1115	0.6655 ± 0.1080	0.6858
CONCH [26] + 2DMamba [60]	**0.6958 ± 0.0367**	0.7317 ± 0.0727	0.7051 ± 0.0486	0.6324 ± 0.1101	0.6860 ± 0.1706	0.6902
E2E CONCH (Ours)	0.6835 ± 0.0152	**0.7421 ± 0.0288**	0.7077 ± 0.0214	0.6412 ± 0.0663	**0.7147 ± 0.0978**	**0.6978**
H0-mini [13] + ABMIL [18]	0.6521 ± 0.1223	0.7012 ± 0.0313	0.6799 ± 0.0555	0.6121 ± 0.1340	0.7210 ± 0.1727	0.6733
H0-mini [13] + CLAM [25]	0.6223 ± 0.0313	0.7169 ± 0.0324	0.6567 ± 0.0751	0.5825 ± 0.0939	0.6394 ± 0.1292	0.6436
H0-mini [13] + DSMIL [22]	0.6549 ± 0.0359	0.7175 ± 0.0347	0.6654 ± 0.0568	0.6020 ± 0.1005	0.6923 ± 0.3163	0.6680
H0-mini [13] + TransMIL [39]	0.6197 ± 0.0616	0.7141 ± 0.0323	0.6447 ± 0.0394	0.5640 ± 0.0535	0.6953 ± 0.1599	0.6495
H0-mini [13] + WiKG [24]	0.6486 ± 0.0466	0.7216 ± 0.0423	0.6868 ± 0.0720	0.5989 ± 0.0891	0.7329 ± 0.2463	0.6778
H0-mini [13] + RRTMIL [41]	0.6525 ± 0.0734	0.7029 ± 0.0344	0.6544 ± 0.0396	0.6596 ± 0.0823	0.7149 ± 0.1605	0.6769
H0-mini [13] + 2DMamba [60]	0.6285 ± 0.0834	0.7073 ± 0.0306	0.6661 ± 0.0612	0.6666 ± 0.1231	0.7366 ± 0.1128	0.6810
E2E H0-mini (Ours)	**0.6581 ± 0.0515**	**0.7239 ± 0.0367**	**0.6925 ± 0.0279**	**0.6869 ± 0.0522**	**0.8176 ± 0.1258**	**0.7158**

Table 1. Comparison results of ViT-Small, CONCH, and H0-mini converted by our proposed E2E-ViT strategies with various two-stage MIL methods on five survival prediction datasets.

3.2.3. Positional Encoding

Positional encoding enables each image token in a ViT to be location-aware, breaking permutation invariance. Most ViT models adopt learnable absolute positional embeddings; however, during sequence extrapolation, positions for additional tokens are obtained via interpolation, which can weaken extrapolation. Therefore, we replace vanilla embeddings with ALiBi [34], a parameter-free relative positional encoding that applies an attention bias based on token distance, thereby improving extrapolation to longer sequences.

4. Experiments

4.1. Datasets and Implementation

We conduct survival prediction experiments over five public cancer datasets from CPTAC [12] and MBC [2]: Clear Cell Renal Cell Carcinoma (CCRCC) (n=218), Head and Neck Squamous Cell Carcinoma (HNSC) (n=243), Lung Adenocarcinoma (LUAD) (n=313), Pancreatic Ductal Adenocarcinoma (PDAC) (n=227), and Metastatic Breast Cancer (MBC) (n=96). We follow [59] to perform a five-fold cross-validation for data partitioning, and report the mean and standard deviation of the concordance index (C-index) to quantify the performance of the models in correctly ranking risk scores according to their overall survival outcomes.

To evaluate our conversion strategy, we conducted experiments with three ViT backbones obtained via distinct pre-training paradigms: ViT-Small [11] (ImageNet pretrained [9]) , CONCH [26] (pathology vision-language pretrained [58]), and H0-mini [13] (pathology image SSL pretrained). For comparison with two-stage MIL methods, since MIL models don't have initialized weights and need to be trained, our converted ViT are evaluated under the full parameter fine-tuning setting. For comparison with SFMs, since these models are already pre-trained, both our converted ViT and SFMs are evaluated under the linear probing setting, where the backbone are frozen and only the classifier is trained. We use Adam optimizer [21] with 10^{-4} learning rate, 1 batch size, and 30 epochs. We adopt early stopping with a patience of 5 consecutive iterations without improvement. Experiments are run on a single NVIDIA A100 80GB GPU and all baseline methods are trained under the same hyperparameters and settings.

Backbone	CCRCC	HNSC	LUAD	PDAC	MBC	Overall
CHIEF [49]	0.5962 ± 0.1147	0.7069 ± 0.0428	0.6778 ± 0.0623	0.6502 ± 0.1838	0.5620 ± 0.2323	0.6386
GigaPath [54]	0.5974 ± 0.0603	0.6475 ± 0.0709	0.6441 ± 0.0559	0.6058 ± 0.1495	0.6196 ± 0.1241	0.6229
MADELEINE [19]	0.5904 ± 0.0525	0.7170 ± 0.0518	0.6831 ± 0.0618	0.5934 ± 0.0444	0.6050 ± 0.1453	0.6378
PRISM [37]	0.6029 ± 0.0893	0.6992 ± 0.0565	0.6883 ± 0.0729	0.6026 ± 0.1040	0.6360 ± 0.2603	0.6458
TITAN [10]	0.5709 ± 0.0509	**0.7186 ± 0.0424**	0.6836 ± 0.0509	0.6556 ± 0.1711	0.6624 ± 0.2013	0.6582
HIPT [5]	0.5190 ± 0.0831	0.6034 ± 0.0364	0.5726 ± 0.0626	0.6264 ± 0.0856	0.6310 ± 0.1055	0.5905
FEATHER [38]	0.5794 ± 0.1351	0.6804 ± 0.0432	**0.6885 ± 0.0735**	0.5809 ± 0.1859	0.5489 ± 0.1873	0.6152
E2E ViT-Small (Ours)	0.5763 ± 0.0340	0.6769 ± 0.0674	0.6872 ± 0.0214	0.5760 ± 0.0605	0.4632 ± 0.1242	0.5959
E2E CONCH (Ours)	0.5941 ± 0.0912	0.7017 ± 0.0755	0.6721 ± 0.0289	0.6417 ± 0.0689	0.6574 ± 0.0816	0.6534
E2E H0-mini (Ours)	**0.6194 ± 0.0941**	0.6983 ± 0.0884	0.6689 ± 0.0589	**0.6604 ± 0.0865**	**0.6954 ± 0.1493**	**0.6685**

Table 2. Comparison results of ViT-Small, CONCH, and H0-mini converted by our proposed E2E-ViT strategies with various SFMs on five survival prediction datasets.

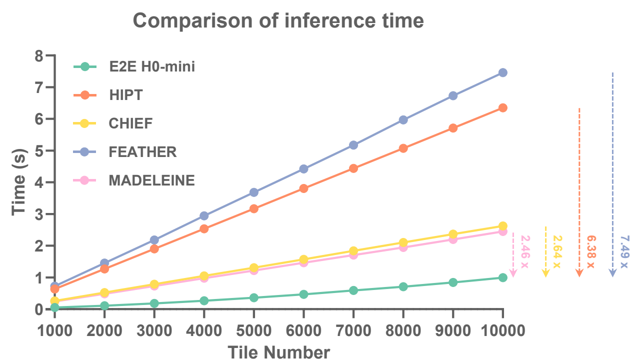

Figure 4. Comparison results for inference time of generating a slide-level representation between our converted E2E H0-mini and four SFMs under different tile image numbers.

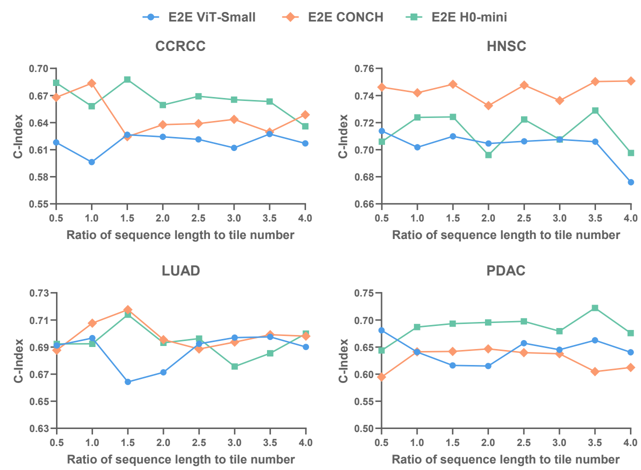

Figure 5. Comparison results of our converted E2E ViT-Small, E2E CONCH, and E2E H0-mini under different ratios of transformer sequence length to tile image number.

4.2. Comparison with Two-Stage MIL

We present our comparative results against two-stage MIL approaches on five datasets in Table 1. We compare E2E-ViT with the following seven state-of-the-art methods: (1) ABMIL [18], an attention-based aggregation framework. (2) CLAM [25], a single-gated attention-based framework with a clustering-constraint loss. (3) DSMIL [22], a dual-stream model employing non-local attention pooling and max pooling. (4) TransMIL [39], a transformer-style method combining linear attention with multiscale positional encoding. (5) WiKG [24], a head-tail dynamic graph network for slide representation. (6) RRTMIL [41], an online re-embedding MIL framework, and (7) 2DMamba [60], a representation model based on two-dimensional Mamba sequence encoding. Across all three converted pre-trained ViTs, our approach achieves consistently superior performance, suggesting that E2E-ViT, by directly "seeing" the raw WSIs, may capture richer information from tissue morphology. Notably, CONCH and H0-mini, which are pre-trained on pathology images, show significant performance improvements under the same MIL methods compared to ViT-Small, which is pre-trained using natural images. However, our E2E ViT-Small, fine-tuned from the raw image view, narrows the gap between ImageNet and pathology pre-trained backbones, demonstrating the promise of end-to-end WSI learning.

4.3. Comparison with SFMs

We further present comparisons against SFMs in Table 2. Specifically, we evaluate our E2E-ViT with five vision-only SFMs: (1) CHIEF [49], which leverages contrastive learning with anatomical priors. (2) GigaPath [54], trained via masked feature reconstruction. (3) MADELEINE [19], which performs contrastive learning across multi-stain WSIs. (4) HIPT [5], composed by dual-scale SSL pre-trained ViT, (5) FEATHER [38], trained with 108-category

Task	Merge	ViT-Small	CONCH	H0-mini
HNSC	Max	**0.7141 ± 0.0540**	0.7387 ± 0.0412	0.7152 ± 0.0277
	Attention	0.7049 ± 0.0580	0.7321 ± 0.0247	0.6931 ± 0.0255
	Mean	0.7018 ± 0.0499	**0.7421 ± 0.0288**	**0.7239 ± 0.0367**
LUAD	Max	0.6929 ± 0.0465	0.6994 ± 0.0346	0.6709 ± 0.0584
	Attention	0.6763 ± 0.0488	0.7018 ± 0.0605	0.6717 ± 0.0686
	Mean	**0.6967 ± 0.0601**	**0.7077 ± 0.0214**	**0.6925 ± 0.0279**
PDAC	Max	0.6147 ± 0.0975	0.6181 ± 0.1316	0.6630 ± 0.0857
	Attention	**0.6481 ± 0.0809**	0.6184 ± 0.1267	**0.6930 ± 0.0734**
	Mean	0.6408 ± 0.0671	**0.6412 ± 0.0663**	0.6869 ± 0.0522

Table 3. Comparison results of our converted E2E ViT-Small, E2E CONCH, and E2E H0-mini with different patch merging methods on HNSC, LUAD, and PDAC datasets.

Task	PE	ViT-Small	CONCH	H0-mini
HNSC	None	0.6918 ± 0.0317	0.6744 ± 0.0069	0.6932 ± 0.0136
	Learnable	**0.7085 ± 0.0150**	0.6920 ± 0.0207	0.6983 ± 0.0319
	ALiBi	0.7018 ± 0.0499	**0.7421 ± 0.0288**	**0.7239 ± 0.0367**
LUAD	None	0.6945 ± 0.0427	0.6350 ± 0.0828	0.6618 ± 0.0306
	Learnable	0.6922 ± 0.0636	0.6765 ± 0.0204	0.6808 ± 0.0239
	ALiBi	**0.6967 ± 0.0601**	**0.7077 ± 0.0214**	**0.6925 ± 0.0279**
PDAC	None	0.6512 ± 0.0931	0.6293 ± 0.0818	0.6747 ± 0.1087
	Learnable	**0.6710 ± 0.0987**	0.6061 ± 0.1397	0.6560 ± 0.0968
	ALiBi	0.6408 ± 0.0671	**0.6412 ± 0.0663**	**0.6869 ± 0.0522**

Table 4. Comparison results of our converted E2E ViT-Small, E2E CONCH, and E2E H0-mini with different position encoding on HNSC, LUAD, and PDAC datasets.

supervised labels with UNI [6] tile features. Additionally, we compare our approach with two vision-language SFMs: (1) PRISM [37], pre-trained under the CoCa framework, and (2) TITAN [10], which employs multi-stage multimodal learning. Our results show that the converted CONCH consistently outperforms the vision-only SFMs and achieves competitive performance compared to two vision-language SFMs. The converted H0-mini overall performs best. We also demonstrate that, although all the above SFMs are trained on pathology images, our converted ViT-Small surpasses several of them across datasets, highlighting the advantages of end-to-end slide representation.

4.4. Ablation Studies

We evaluate the feature extraction efficiency of our E2E H0-mini compared with existing SFMs under different numbers of cropped tiles, as illustrated in Figure 4. We observe that E2E H0-mini can output features within one second for different tile numbers, achieving the fastest inference speed. For example, it is 2.64 times faster than CHIEF and 7.49 times faster than FEATHER with 10,000 tile numbers.

We further examine the impact of the input sequence length on the transformer backbone. Based on the equation (2), we conduct experiments with fine-tuning settings under different ratios of sequence length to tile number, as shown in Figure 5. The results vary across datasets, suggesting that different tumor types may require distinct receptive field sizes to effectively capture morphological patterns. Nevertheless, the overall performance remains relatively stable across sequence lengths, indicating that the pre-trained transformer backbone possesses a strong extrapolation capability.

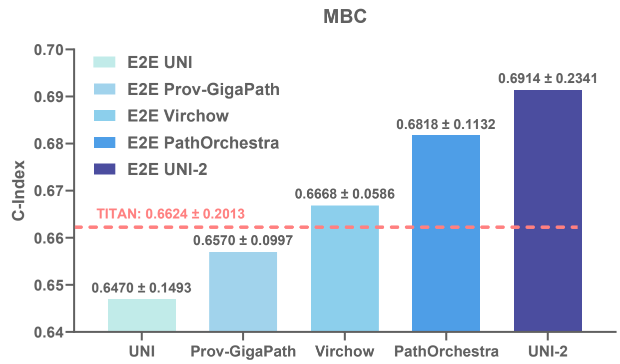

Figure 6. Comparison results of five larger pre-trained ViTs converted by our proposed E2E-ViT strategies under linear probe settings on the MBC dataset.

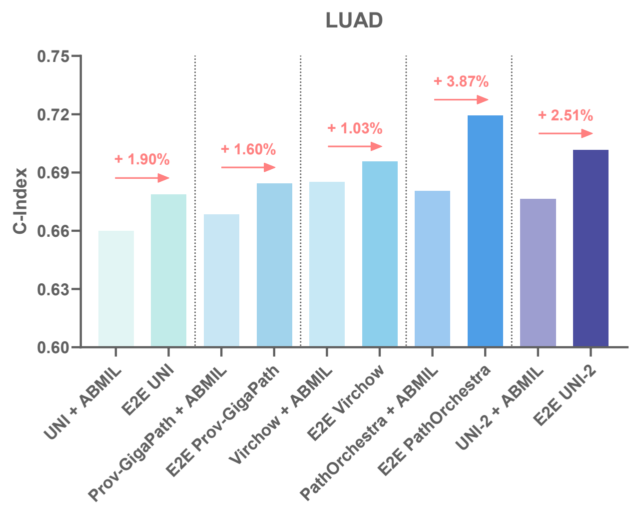

Figure 7. Comparison results between five larger pre-trained ViT converted by our proposed E2E-ViT strategies and the two-stage ABMIL method under fine-tuning settings on the LUAD dataset.

We also evaluate different patch merging strategies under fine-tuning settings, as reported in Table 3. We found that the mean pooling over all patches consistently outperforms the non-parameter max pooling. Notably, attention pooling achieves comparable performance. However, it requires additional training, thereby compromising the plug-and-play nature of the converted ViT for directly generating slide representations. We will explore learnable patch merging strategies in future work. Furthermore, Table 4 presents an ablation on positional encoding schemes under fine-tuning settings, revealing that ViT models equipped with relative positional encoding demonstrate improved adaptability for WSI representation.

4.5. Experiments on Larger ViTs

Recent large-sized pre-trained ViTs have been widely adopted as tile-level encoders in two-stage MIL and in SFM pipelines. We further examine the capability of converting five large ViTs pre-trained on pathology images: UNI [6], Prov-GigaPath [54], Virchow [44], PathOrchestra [56], and UNI-2 [6]. First, all five converted ViTs outperform the two-stage ABMIL baseline on LUAD datasets. Second, under the linear probing setting with frozen backbones, the large converted ViTs retain advantages. For example, Virchow, PathOrchestra, and UNI-2 surpass TITAN on MBC. It is worth noting that although converting these large ViTs into end-to-end WSI models is feasible, the computational cost is substantial, and their tightly coupled parameters make them sensitive to perturbations, which may result in the loss of pre-trained prior knowledge. Therefore, a careful trade-off between computational efficiency and performance should be considered when applying such models.

4.6. Visualization

We also visualize the CLS token attention of the converted E2E CONCH as heatmaps, as shown in Figure 8. We observe that E2E CONCH concentrates attention effectively in cancerous regions, indicating its ability to capture disease-specific morphological cues. Moreover, the end-to-end model exhibits a finer attention distribution, characterized by more diverse red-scale transitions and a greater number of tissue boundaries. In contrast, SFMs tend to produce highly saturated hotspots over localized areas. These patterns suggest that the end-to-end model may distribute attention more evenly across the slide and better differentiate region-level uniqueness, which could be advantageous for tasks such as survival analysis that are sensitive to global spatial context.

5. Conclusion and future work

In this work, we dissect the ViT architecture and propose a conversion strategy called E2E-ViT based on the sequence extrapolation capability of ViTs in pathology images. E2E-ViT can transform ViTs pre-trained on low-resolution tiles into models capable of end-to-end high-resolution WSI representation without adding trainable parameters. E2E-ViT introduces a redesigned image input, an input sequence compression mechanism for the Transformer backbone, and a relative positional encoding. Across multiple survival prediction tasks, models converted by E2E-ViT surpass representative two-stage MIL pipelines and SFMs. Our approach presents a new paradigm for WSI analysis that effectively addresses prior limitations, opening a promising avenue for next-generation WSI learning.

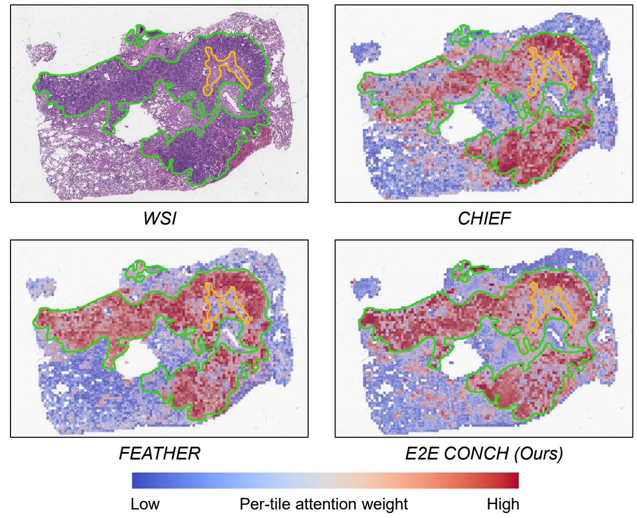

Figure 8. Attention heatmap visualization of SFMs and our converted E2E CONCH.

Our future work plans to strengthen long-sequence extrapolation by post-training on high-resolution images, incorporate the multi-scale mechanism into the end-to-end model, and develop a vision–language variant to explore multimodal understanding. We will also validate the models on more diverse clinical pathology tasks.

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